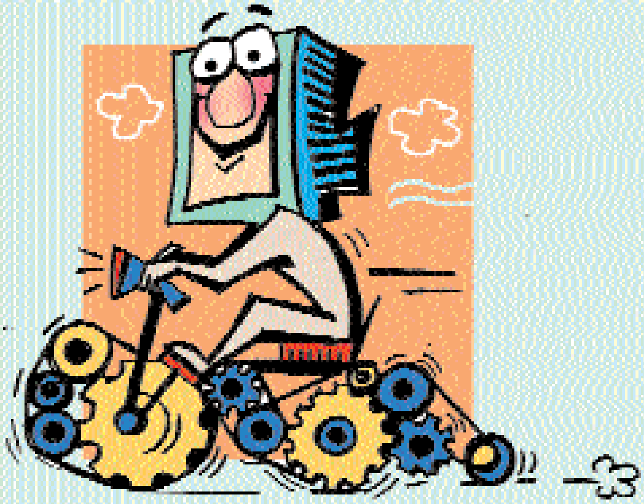
IN THE FAST LANE

Fix the BIOS settings to shed precious seconds from boot time. Fine tune CPU settings for superior system performance



Reducing boot time

Minimise the time it takes for your computer to boot by changing a few basic settings. Enter the BIOS by holding down the [Delete] key once you switch on your computer and go to the Advanced BIOS options of the main BIOS menu.

Begin by ensuring that the Quick POST (Power-On Self Test) is enabled. If the option is available, disable the selection that performs a floppy drive seek during boot up. Furthermore, in the same Advanced BIOS screen, set your first boot device as your hard disk, or the system will waste several moments in attempting to boot from another device such as the floppy, CD-ROM or Zip drive.

These steps alone can cast off up to 15 seconds of boot time.

Boosting memory speed

Being one of the major bottlenecks in overall performance, a slight boost in memory speed can have a deciding effect on how peppy your PC is. If you have 133 MHz (or faster) modules installed in a relatively modern motherboard, you can adjust the system memory frequency accordingly. The PC133 specification delivers 33 per cent more throughput over the PC100 variety, which comes in useful with regard to memory demanding programs, especially games. More often than not, *Quake III* shows at least a 15 per cent rise in fps just by using quicker memory.

You can squeeze your system's memory for more juice by lowering the time it takes to respond. Usually, system memory takes about three cycles to react. However, most memory modules can do it quicker. Let's try bringing this value down further. Proceed to the Advanced Chipset option and select the entry that reads as SDRAM CAS Latency Time. Bring the value down by a notch from 3 to 2. While inside the same set of options, look for SDRAM RAS-to-CAS Delay and SDRAM RAS Precharge Time and change those to two as well.

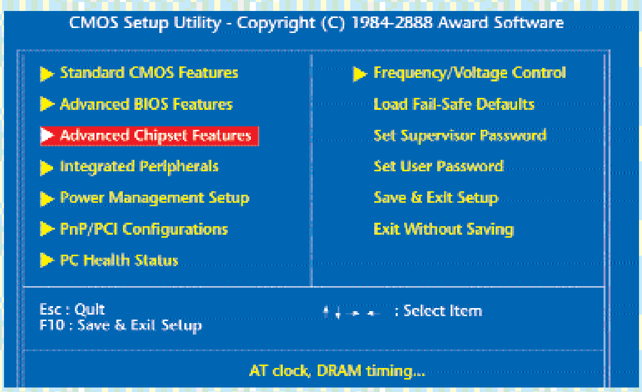
The final scene to conquer on the memory front comes with adjusting how much time it takes for data transactions to happen. The default value for this is 6/8, which is slower, albeit more stable than lower values. Lowering the number will boost system performance, but at the risk of stability. It's usually possible to bring the SDRAM Cycle Time down to 5/7; your hardware should be able to handle it. Look for this particular entry in the Advanced Chipset option.

Cache is the key

A cache, in computer terms, is nothing more than a digital scratchpad used for storing frequently accessed information. It works the same way as memory does, but its lightning fast speed set it so apart, they had to change its name. A CPU sports a dedicated cache where it jots down its calculations. If you take away its cache, then the CPU reverts to RAM, with a performance hit so incredible you'll undergo a first hand feel of its significance.

System memory also moonlights as a cache. It's slower than CPU cache, but faster than BIOS firmware. Enabling system and video BIOS caching copies the firmware into the main memory, where it can be accessed rapidly.

As with many of the tweaks given here, these two options are present in the



Press [Delete] at system startup; it should get you into the blue world of the BIOS